

Growth Rate Response & YMC

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“Whether we see it or not, we know that deep in their hearts they are oscillating!”

Sandy Silverman

Comparing Slopes in Glucose and Ethanol

¹Brauer *at.al.* (2007)

Comparing Slopes in Glucose and Ethanol

- Growth rates in Glu¹: $\underbrace{\{0.05, 0.10, 0.15\}}_{s_1}, \underbrace{\{0.20, 0.25, 0.30\}}_{s_2} h^{-1}$

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- Correlations using all 6 growth rates, s_1 & s_2 :

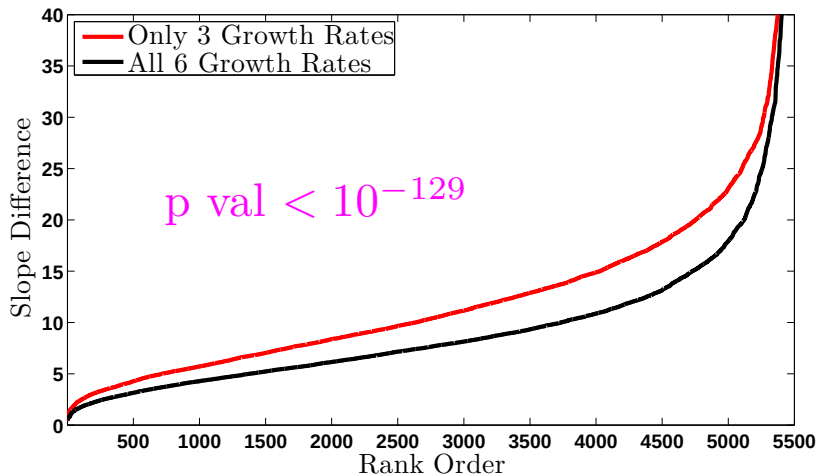
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- Correlations using the 3 corresponding growth rates, s_1 :
 - Carbon Source: 0.10
 - Nitrogen: 0.26
 - Phosphate: 0.15

Difference between Slopes in Glucose and Ethanol



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At steady-state, the measured level of the i^{th} gene ($\hat{G}_i$) is:

$$\hat{G}_i = \frac{1}{T} \int_{t=0}^{t=T} G(t) dt \approx \frac{1}{\sum_{j \in \omega} T_j} \sum_{j \in \omega} T_j G_{ij} \quad (1)$$

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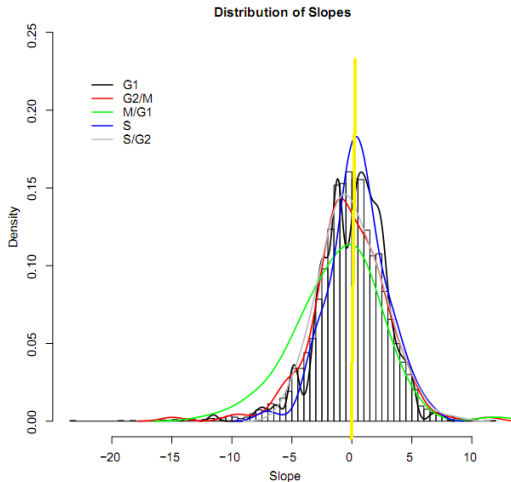
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- $\mu = 0.14h^{-1} \mapsto (T_S + T_{G2} + T_M)/T_{G1} = 0.28$
- $\mathbf{E}[Slope]$ for $S/G2/M$ $\log_2(0.28/0.15)/(0.14 - 0.05) = \mathbf{10}$

Distribution of Slopes for the Cell Cycle Genes



Brauer *at.al.* (2007) & Spellman *at.al.* (1998)

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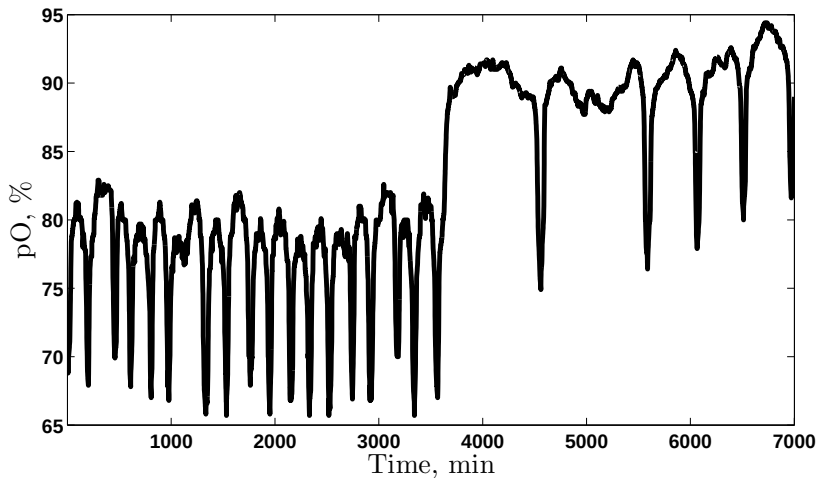
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 - FISH in non synchronized cultures

Growth Rate effect on the YMC



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